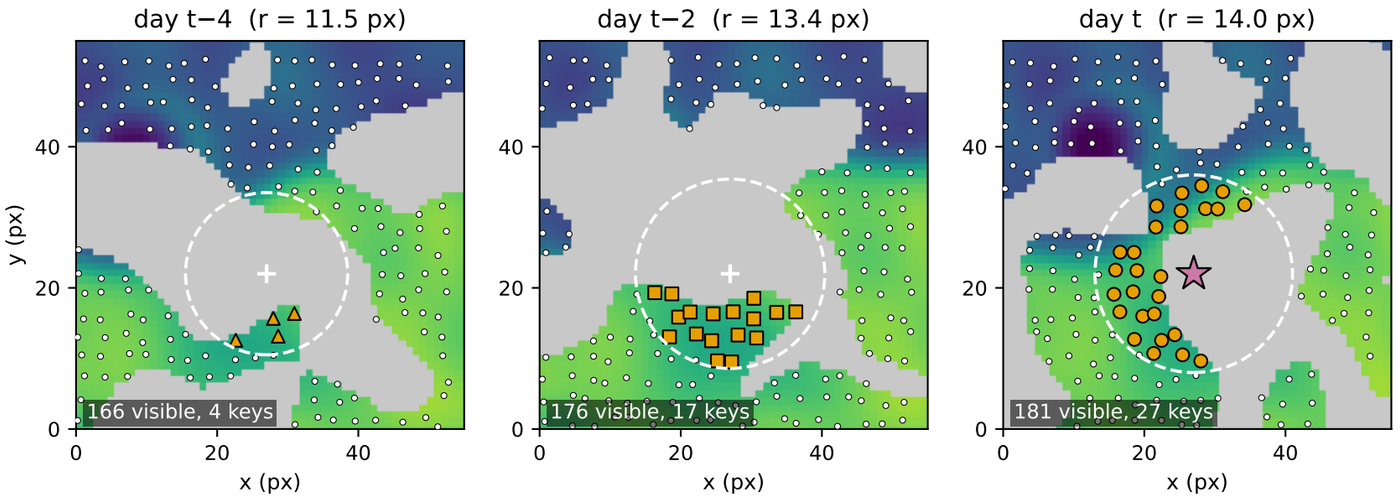
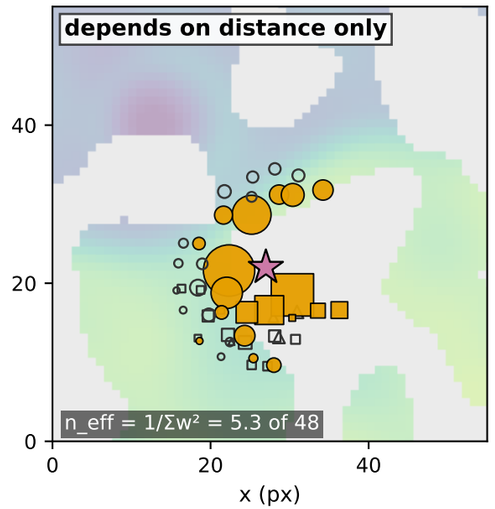
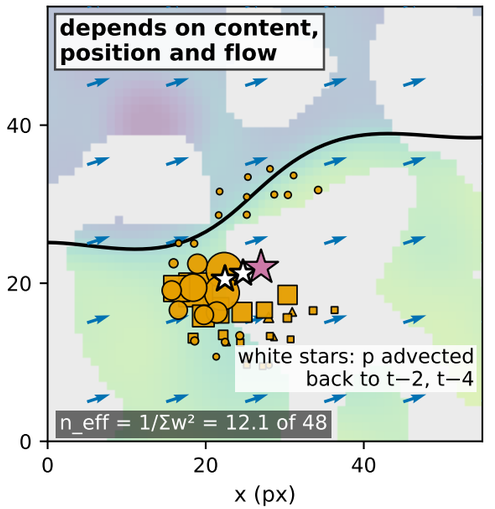
A. Setting: visible observations around a cloudy target pixel, three of the ±8 days



B. Classic kriging weights



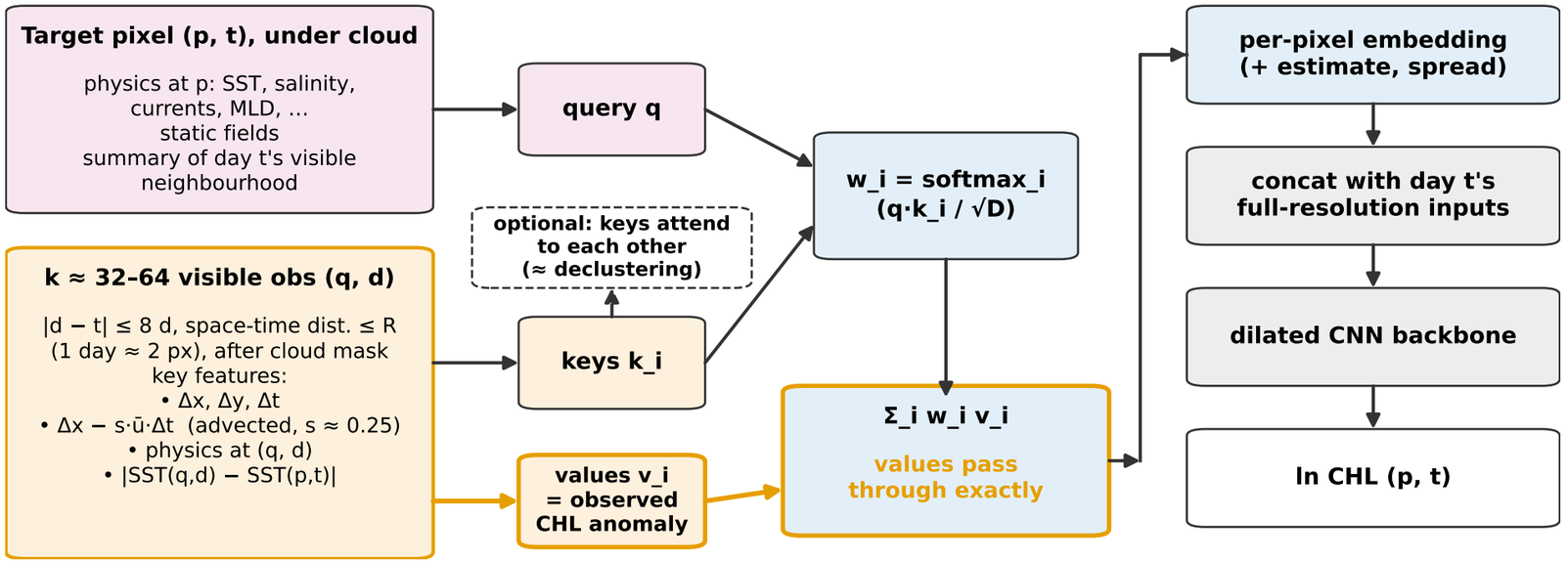
C. Learned kriging weights (toy rule)





target pixel p (day t) · visible obs. (display subsample) · selected key, day t · key, day t−2 · key, day t−4 · radius R in space-time (1 d = 2 px) · cloud · SST front (C) · current ū (C) · marker size = weight; hollow = w < 0

D. Learned kriging: architecture for one target pixel (p, t)



E. Context sources by model

	same-day neighb.	own px, other days	nearby px, other days	motion-aware	exact values
arm A (own value)	CNN	yes	4×4 summary	no	own px
arm B (Gauss. avg)	CNN	blurred	blurred σ 1.5 px	no	no
per-pixel attention	CNN	embedded	3×3 conv embedded	no	no
tokens (4×4 patches)	CNN	4×4 summary	4×4 summary	no	no
learned kriging	keys Δt = 0	yes	yes, per obs.	advected offset	yes

CNN: day t's own pixels via the backbone (all models).
Cells: green = full, orange = partial, grey = none.